

Lesson 26: The Diffusion Equation - Dimensional Analysis & Self-Similarity

Professor Dave Explains Differential Equations Master Series · Lesson 26

Video: The Diffusion Equation Part 2: Dimensional Analysis and Self-Similarity Course: Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that on an infinite line ($-\infty < x < \infty$), there is no physical length scale L . Without an L , Fourier separation of variables breaks down! How can we solve it? Through Dimensional Analysis! Thermal diffusivity α has units m^2/s . To combine position x [m] and time t [s] into a dimensionless variable, nature demands the Similarity Variable $\eta = x / \sqrt{4\alpha t}$. This remarkable transformation collapses the 2D partial differential equation into a single 1D ordinary differential equation, giving birth to the Error Function $\text{erf}(z)$.

2. Core Theoretical Framework & Rigorous Formulations

1. The Similarity Transformation Engine

For $u_t = Du_{xx}$ on a semi-infinite rod ($x > 0$), let:

$$\eta = \frac{x}{\sqrt{4Dt}}, \quad u(x, t) = f(\eta)$$

By the chain rule: $\frac{\partial u}{\partial t} = -\frac{\eta}{2t} f'(\eta)$ and $\frac{\partial^2 u}{\partial x^2} = \frac{1}{4Dt} f''(\eta)$. Substituting collapses the PDE into the ODE:

$$f''(\eta) + 2\eta f'(\eta) = 0$$

2. Integration to the Error Function

$$\frac{f''}{f'} = -2\eta \implies \ln(f') = -\eta^2 + C_1 \implies f'(\eta) = Ae^{-\eta^2}$$

$$f(\eta) = A \int_0^\eta e^{-s^2} ds + B$$

The Gauss Error Function: $\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-s^2} ds$, $\text{erfc}(z) = 1 - \text{erf}(z)$.

3. Fully Solved Whiteboard Problems

Materials Science Connection: Carbon Carburization in Steel (Callister Eq 5.5)

Problem Statement: Solve $u_t = Du_{xx}$ for $x > 0, t > 0$ subject to initial concentration $u(x, 0) = C_0$ and constant surface boundary concentration $u(0, t) = C_s$.

Step 1: Formulate Boundary Conditions in Similarity Variable η

At $x = 0: \eta = 0 \implies f(0) = C_s$. As $x \rightarrow \infty$ (or $t \rightarrow 0$): $\eta \rightarrow \infty \implies f(\infty) = C_0$.

Step 2: Apply General Error Function Solution

$f(\eta) = A \int_0^\eta e^{-s^2} ds + B$. At $\eta = 0: f(0) = 0 + B = C_s \implies B = C_s$.

Step 3: Enforce Far-Field Asymptotic Condition

As $\eta \rightarrow \infty: f(\infty) = A \int_0^\infty e^{-s^2} ds + C_s = A \frac{\sqrt{\pi}}{2} + C_s = C_0 \implies A = \frac{2(C_0 - C_s)}{\sqrt{\pi}}$.

Step 4: Substitute and Group into Error Function

$u(x, t) = \frac{2(C_0 - C_s)}{\sqrt{\pi}} \int_0^\eta e^{-s^2} ds + C_s = (C_0 - C_s)\text{erf}(\eta) + C_s = C_s - (C_s - C_0)\text{erf}(\eta)$.

Step 5: Canonical Engineering Form

$\frac{u(x, t) - C_0}{C_s - C_0} = 1 - \text{erf}\left(\frac{x}{\sqrt{4Dt}}\right) = \text{erfc}\left(\frac{x}{\sqrt{4Dt}}\right)$. Exactly Callister Equation 5.5 in materials science for case-hardening of gears!

⚠ Common Exam Pitfall & Trapdoor Avoidance

Remember that $\text{erf}(0) = 0$ and $\text{erf}(\infty) = 1$. Mixing up the limits or failing to normalize by $2/\sqrt{\pi}$ causes amplitude scaling errors.

4. Key Takeaways & Exam Strategy Formula Card

 Core Takeaway & Exam Rule

Dimensional analysis collapses infinite-domain diffusion into $\eta = x / \sqrt{4Dt}$, yielding the universal Error Function $\text{erfc}(\eta)$.

- Video reference: [The Diffusion Equation Part 2: Dimensional Analysis and Self-Similarity](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.